

Reddy Harsha Vardhan

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SUMMARY

Software Engineer with 4+ years of experience designing and developing scalable, cloud-native web applications. Proficient in full-stack development, including frontend frameworks such as React.js, Next.js, TypeScript, and Tailwind CSS, alongside backend expertise in Java (Spring Boot, Hibernate), Node.js, and microservices within Agile environments. Experienced in deploying and managing applications on AWS using Docker, Kubernetes, Terraform, and CI/CD pipelines (GitHub Actions, Jenkins, GitLab). Skilled in database architecture and optimization across SQL and NoSQL databases (PostgreSQL, MySQL, MongoDB, DynamoDB, Cassandra), with experience in event-driven architectures leveraging Kafka. Strong knowledge of application security with OAuth2, JWT, and Spring Security, and integrating Generative AI and LLM solutions to enhance automation, intelligence, and user experience.

EDUCATION

Master of Science in Computer Science | Saint Louis University

Saint Louis, Missouri

TECHNICAL SKILLS

Programming & Frameworks: Java, Spring Boot, Hibernate, Python, Node.js, Express.js, RESTful APIs, Microservices, GraphQL, TypeScript, Angular.js, Next.js
Frontend & UI: React.js, Redux Toolkit, Angular.js, TypeScript, HTML5, CSS3, Tailwind CSS, Responsive & Component-based Design
Cloud & DevOps: AWS (EC2, S3, Lambda, DynamoDB, RDS, API Gateway, EKS, Elastic Beanstalk), Docker, Kubernetes, Terraform, Jenkins, GitHub Actions, GitLab CI/CD, Kafka, Prometheus, Grafana, CloudWatch
Databases & Caching: PostgreSQL, MySQL, MongoDB, Redis, Cassandra, SQL Optimization, Data Modeling
System Design & Architecture: Distributed Systems, API Design, Microservices Architecture, High Availability, Concurrency & Parallelism
Security & API Management: OAuth2, JWT, Spring Security, RBAC, API Gateway, Swagger, Postman
Testing & Performance: JUnit, Mockito, Jest, Mocha, Cypress, Selenium, Load Testing, Performance Tuning, CI/CD Integration
AI & Generative Models: LLMs (BERT, GPT, RoBERTa), LangChain, OpenAI GPT API, Generative AI, Prompt Engineering
Agile & Collaboration: Scrum/Kanban, JIRA, Confluence, Code Reviews, Sprint Planning, Cross-functional Teamwork

PROFESSIONAL EXPERIENCE

Paypal September 2024 – Present | United States
Software Engineer

- Improved reliability of core payment services by enhancing **Java and Spring Boot microservices** with concurrency-safe execution, idempotent APIs, and asynchronous processing, reducing duplicate transaction handling by **28%** and preventing inconsistent state propagation.
- Reduced latency in distributed payment workflows by integrating **Apache Kafka** with partitioned topics, controlled retries, and dead-letter handling, improving transaction throughput by **15,000 events per minute** during peak load.
- Supported cloud infrastructure on **AWS (EC2, ECS, Lambda, S3, DynamoDB)** and contributed to **Terraform-based provisioning**, standardizing environments and improving deployment reliability across staging and production.
- Boosted observability by implementing **CloudWatch, Prometheus, and Grafana dashboards** with real-time alerts, log aggregation, and service-level indicators, increasing anomaly detection coverage by **2.5X** compared to previous monitoring.
- Refined system resilience by applying **circuit breakers, rate limiting, and graceful degradation**, reducing user-facing payment errors from **120 incidents per month to 45** during dependency outages.
- Configured containerized deployments using **Docker and Kubernetes**, implementing rolling updates and auto-scaling, which increased deployment success rate to **>99%** and minimized rollback frequency.
- Participated in **API and system design reviews**, contributing to request validation, pagination, backward compatibility, and distributed data consistency, improving long-term maintainability.
- Enhanced user-facing experiences by developing **React-based components with Redux Toolkit**, optimizing render efficiency and cutting memory usage by **20%**, improving overall web platform responsiveness.
- Fine - tuned release quality by contributing to **CI/CD pipelines**, automated testing with **JUnit and Mockito**, and structured code reviews, reducing post-deployment defects by **25%**.

LTIMindtree October 2020 – July 2023 | India
Software Engineer

- Developed and maintained **Java-based backend services** using **Spring Boot and REST APIs** for a high-performance financial platform, increasing request handling capacity from 1200 to 1650 requests per minute and supporting concurrent integrations with downstream systems.
- Delivered **full stack functionality** using **Java, Spring MVC, React, and REST APIs**, reducing feature turnaround time by 30% through reusable UI components, standardized service contracts, and consistent error-handling across modules.
- Designed and implemented an **internal monitoring dashboard** using **React and Node.js**, reducing manual service health checks and feature flag verifications by 40%, while providing real-time visibility and alerts for engineering teams.
- Optimized **SQL queries, indexing strategies, and schema usage**, improving application responsiveness and efficiency during concurrent workloads with high query volumes.
- Established and maintained CI/CD pipelines with Jenkins, Docker, and Git, improving deployment reliability and decreasing rollback incidents by 2X.
- Provisioned cloud deployments on **AWS (EC2, Elastic Beanstalk, S3, RDS)**, performing resource right-sizing, storage optimization, and automated scaling, contributing to approximately **\$450K in infrastructure cost savings** over multiple releases while maintaining high availability.
- Implemented **Redis caching and MongoDB-based NoSQL storage** for frequently accessed transaction and user session data, reducing database read load and improving overall API response times under concurrent workloads.
- Strengthened platform security using **Spring Security, OAuth2, JWT, and role-based access controls**, ensuring secure service-to-service and user-level authorization, auditing authentication logs, and maintaining sub-150ms average API response times.
- Collaborated in **Agile development cycles**, actively participating in sprint planning, backlog grooming, code reviews, and defect triage, reducing defect resolution time by 35% and improving sprint delivery predictability and team velocity.

PROJECTS

Smart Banking Query Assistant Java Spring Boot, React, Redux Toolkit, PostgreSQL, Redis, LangChain, OpenAI GPT API, Docker, Kubernetes, AWS

- Constructed an **AI-powered banking assistant** using **LangChain** and **GPT APIs**, enabling multi-turn conversational queries for account balances, transaction history, and payment instructions, handling **80 concurrent simulated user sessions at same time**.
- Built a **React + Redux frontend** integrated with **Spring Boot backend** and **PostgreSQL**, leveraging **Redis** caching for efficient session retrieval, ensuring smooth response times and seamless user experience during simulations.

Real-Time Portfolio Management Platform Java Spring Boot, React, TypeScript, MongoDB, Redis, OpenAPI, Docker, AWS

- Programmed a **portfolio management SaaS platform** integrating **OpenAPI** feeds for real-time stock, mutual fund, and crypto tracking, enabling users to monitor up to **500 simulated assets per portfolio** with sub-2-second dashboard update latency.
- Applied backend services with **Spring Boot** and **MongoDB**, while **Redis** caching and asynchronous processing ensured fast portfolio data updates to **React-based dashboards**, improving **simulated user workflow efficiency** and providing real-time insights for portfolio adjustments.